Long-term effects of lower versus higher oxygenation levels in adult ICU patients – a systematic review

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Electronic Supplementary Material

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PRISMA Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 4 + ESM
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	ESM
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 4 + ESM
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5 + Pages 7-9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	ESM
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Pages 4 + ESM

Section and Topic	Item #	Checklist item	Location where item is reported
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 5
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 5
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Pages 4-5 + ESM
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 6
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 7 + Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Page 7+ Table 1 + ESM
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 + ESM
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Pages 8-9; Table 2 + Figure 2

Section and Topic	Item #	Checklist item	Location where item is reported
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Pages 8-9
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Pages 8-9; Figures 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Pages 8-9+ ESM
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 9
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	ESM
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Pages 8-9 + Table 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pages 10-11
	23b	Discuss any limitations of the evidence included in the review.	Pages 10-11
	23c	Discuss any limitations of the review processes used.	Pages 10-11
	23d	Discuss implications of the results for practice, policy, and future research.	Pages 10-11
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Pages 3-4
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 4
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 11
Competing interests	26	Declare any competing interests of review authors.	Page 11
Availability of	27	Report which of the following are publicly available and where they can be found: template data collection forms; data	Page 4 +

Section and Topic	Item #	Checklist item	Location where item is reported
data, code and other materials		extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	ESM

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-analysis[1];ESM: Electronic Supplementary Material

Search strategy

Cochrane Library – 06.01.2022 – 16.409 hits

Search history link:

https://www.cochranelibrary.com/advanced-search/search-manager?search=6890628

Search history details:

#1	MeSH descriptor: [Hyperoxia] explode all trees	220
#2	MeSH descriptor: [Hypoxia] explode all trees	2241
#3	MeSH descriptor: [Oxygen Inhalation Therapy] explode all trees	1681
#4	MeSH descriptor: [Oxygen] explode all trees	5713
#5	(inspir* or inhal* or fraction* or concentrat* or arterial* or saturation or level* or tension* or supply* or supplement* or supplie* or therap* or administr* or dosag* or dose* or dosing*) near/3 (oxygen):ti,ab,kw	24992
#6	(hyperoxia or hyperoxemia or hyperoxaemia or hypoxia or hypoxemia or hypoxaemia or anoxia or anoxemia or anoxaemia or "arterial oxygen" or "high oxygen" or oxygenat* or "blood gas" or "oxygen saturation" or pao2 or sao2 or spo2 or fio2):ti,ab,kw	34879
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#8	MeSH descriptor: [Critical Illness] explode all trees	2521
#9	MeSH descriptor: [Critical Care] explode all trees	2190
#10	MeSH descriptor: [Intensive Care Units] explode all trees	3963
#11	MeSH descriptor: [Emergency Medicine] explode all trees	290
#12	MeSH descriptor: [Emergency Service, Hospital] explode all trees	2677
#13	((emergency next department*) or ED or (emergency next room*) or ER or (high next dependency next unit*) or HDU or prehospital* or pre-hospital* or "critically ill" or "acutely ill" or "intensive care" or "critical care" or ICU* or (coronary next care next unit*) or (neurological next intermediate next care next unit*)):ti,ab,kw	64943
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#15	MeSH descriptor: [Myocardial Ischemia] explode all trees	29497
#16	("cardiac arrest" or (cardiac next failure*) or CPR or "heart arrest" or (heart next failure*) or (myocardial next infarct*) or "myocardial ischemia" or "acute coronary syndrome"):ti,ab,kw	72748
#17	MeSH descriptor: [Shock] explode all trees	2457
#18	(shock):ti,ab,kw	11654
#19	MeSH descriptor: [Meningitis] explode all trees	711

#20	(meningitis):ti,ab,kw	2012
#21	MeSH descriptor: [Pneumonia] explode all trees	4997
#22	(pneumonia):ti,ab,kw	17890
#23	MeSH descriptor: [Pulmonary Disease, Chronic Obstructive] explode all trees	6174
#24	(COPD or "chronic obstructive pulmonary disease"):ti,ab,kw	20394
#25	MeSH descriptor: [Acute Lung Injury] explode all trees	555
#26	((acute next lung next injur*) or ALI):ti,ab,kw	3080
#27	MeSH descriptor: [Respiratory Distress Syndrome] explode all trees	2623
#28	((adult or acute) near/2 (Respiratory Distress)):ti,ab,kw	7703
#29	MeSH descriptor: [Pulmonary Embolism] explode all trees	1086
#30	((pulmonary or lung) near/2 (embolism or infarct*)):ti,ab,kw	5366
#31	MeSH descriptor: [Multiple Trauma] explode all trees	244
#32	("severe trauma" or "multiple trauma"):ti,ab,kw	799
#33	MeSH descriptor: [Craniocerebral Trauma] explode all trees	3938
#34	((traumatic next brain next injur*) or TBI or "head trauma" or "craniocerebral trauma"):ti,ab,kw	5530
#35	MeSH descriptor: [Stroke] explode all trees	10955
#36	(stroke):ti,ab,kw	60269
#37	MeSH descriptor: [Sepsis] explode all trees	4810
#38	MeSH descriptor: [Shock, Septic] explode all trees	1048
#39	(sepsis or "septic shock"):ti,ab,kw	13573
#40	MeSH descriptor: [Intracranial Hemorrhages] explode all trees	2098
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#42	MeSH descriptor: [Poisoning] explode all trees	2211
#43	(severe near/2 (poisoning or intoxication*)):ti,ab,kw	69
#44	MeSH descriptor: [Diabetic Ketoacidosis] explode all trees	149
#45	("diabetic ketoacidosis"):ti,ab,kw	674
#46	MeSH descriptor: [Liver Failure, Acute] explode all trees	147

#47	((acute next hepatic next failur*) or (fulminating next hepatic next failur*) or (acute next liver next failure*)):ti,ab,kw	249
#48	MeSH descriptor: [Acute Kidney Injury] explode all trees	1596
#49	((acute next kidney next failur*) or (acute next renal next injur*)):ti,ab,kw	2693
#50	MeSH descriptor: [Intestinal Perforation] explode all trees	151
#51	MeSH descriptor: [Appendicitis] explode all trees	625
#52	((intestinal next perforation*) or appendicitis):ti,ab,kw	1885
#53	((acute or emergency) near/2 (surgery or surgical or operat* or resection*)):ti,ab,kw	2908
#54	MeSH descriptor: [Coronavirus] explode all trees	640
#55	MeSH descriptor: [Coronavirus Infections] explode all trees	1587
#56	MeSH descriptor: [COVID-19] explode all trees	1042
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#58	(#8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35 or #36 or #37 or #38 or #39 or #40 or #41 or #42 or #43 or #44 or #45 or #46 or #47 or #48 or #49 or #50 or #51 or #52 or #53 or #54 or #55 or #56 or #57)	265349
#59	(#7 and #58)	16640
#60	#59 in Trials	16409

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7	1 or 2 or 3 or 4 or 5 or 6	488543
8	exp Critical Illness/	34631
9	exp Critical Care/	63117
10	exp Intensive Care Units/	96929
11	exp Emergency Medicine/	14894
12	exp Emergency Service, Hospital/	90934
13	(emergency department* or ED or emergency room* or ER or high dependency unit* or HDU or prehospital* or pre-hospital* or critical illness or critically ill or acutely ill or intensive care or critical care or ICU* or coronary care unit* or neurological intermediate care unit*).tw.	510924
14	exp Heart Arrest/	52596
15	exp Myocardial Ischemia/	452228
16	(cardiac arrest or cardiac failure or CPR or heart arrest or heart failure or myocardial infarct* or myocardial ischemia or acute coronary syndrome).tw.	453337
17	exp Shock/	81132
18	shock.tw.	189980
19	exp Meningitis/	57368
20	meningitis.tw.	54926

21	exp Pneumonia/	226550
22	pneumonia.tw.	133219
23	exp Pulmonary Disease, Chronic Obstructive/	61795
24	(COPD or chronic obstructive pulmonary disease).tw.	70326
25	exp Acute Lung Injury/	7377
26	(acute lung injury or ALI).tw.	17890
27	exp Respiratory Distress Syndrome, Adult/	37557
28	(((adult or acute) adj2 respiratory distress) or ARDS).tw.	27862
29	exp Pulmonary Embolism/	41419
30	((pulmonary or lung) adj (embolism or infarct*)).tw.	37262
31	exp Multiple Trauma/	13306
32	(severe trauma or multiple trauma).tw.	6392
33	exp Craniocerebral Trauma/	169203
34	(traumatic brain injury or TBI or head trauma or craniocerebral trauma).tw.	54485
35	exp Stroke/	153401
36	stroke.tw.	269204
37	exp Sepsis/	132858
38	exp Shock, Septic/	23900
39	(sepsis or septic shock).tw.	120888
40	exp Intracranial Hemorrhages/	75857
41	((intracranial or intra-cranial or subarachnoid or cerebral or life-threatening) adj2 (hemorrhage or haemorrhage or bleeding)).tw.	49904
42	exp Poisoning/	163928
43	(severe adj (poisoning or intoxication*)).tw.	945
44	exp Diabetic Ketoacidosis/	6970
45	diabetic ketoacidosis.tw.	6376
46	exp Liver Failure, Acute/	6922
47	(acute hepatic failure* or fulminating hepatic failure* or acute liver failure*).tw.	7941
48	exp Acute Kidney Injury/	51247

49	(acute kidney failure* or acute renal injur*).tw.	2186
50	exp Intestinal Perforation/	13572
51	exp Appendicitis/	20151
52	(intestinal perforation* or appendicitis).tw.	23109
53	((acute or emergency) adj2 (surgery or surgical or operat* or resection*)).tw.	30796
54	exp Coronavirus/	116062
55	exp Coronavirus Infections/	141102
56	exp COVID-19/	130314
57	(coronavirus or corona virus or covid or covid2019 or covid19 or SARS2 or SARS-CoV-2 or SARS-CoV-19 or severe acute respiratory pneumonia outbreak or novel cov or 2019ncov or sars cov2 or cov22 or ncov or coronaviridae).tw.	218032
58	8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57	2881395
59	7 and 58	105289
60	randomized controlled trial.pt.	555133
61	controlled clinical trial.pt.	94628
62	randomi?ed.ab.	653072
63	placebo.ab.	224541
64	clinical trial.sh.	533117
65	randomly.ab.	373388
66	trial.ti.	254201
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68	exp animals/ not humans.sh.	4939103
69	67 not 68	1428445
70	59 and 69	10634

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4	*oxygen/	57589
5	*arterial oxygen saturation/	1045
6	*oxygen blood level/	633
7	*arterial oxygen tension/	2388
8	*blood oxygen tension/	700
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12	*critical illness/	12972
13	*intensive care/	64548
14	*intensive care unit/	38384
15	*emergency medicine/	28270
16	*emergency health service/	49118
17	*coronary care unit/	3762
18	(emergency department* or ED or emergency room* or ER or high dependency unit* or HDU or prehospital* or pre-hospital* or critical illness or critically ill or acutely ill or intensive care or critical care or ICU* or coronary care unit* or neurological intermediate care unit*).tw.	795996
19	*heart arrest/	25693
20	*acute heart infarction/	42958

21	(cardiac arrest or cardiac failure* or CPR or heart arrest or heart failure* or myocardial infarct* or myocardial ischemia or acute coronary syndrome).tw.	691750
22	*shock/	11253
23	shock.tw.	242141
24	*meningitis/	14828
25	meningitis.tw.	61133
26	*pneumonia/	46379
27	pneumonia.tw.	193378
28	*chronic obstructive lung disease/	74808
29	*acute lung injury/	8002
30	(acute lung injur* or ALI).tw.	27875
31	(COPD or chronic obstructive pulmonary disease).tw.	122754
32	*adult respiratory distress syndrome/	17784
33	(((adult or acute) adj2 respiratory distress) or ARDS).tw.	40996
34	*lung embolism/	35455
35	((pulmonary or lung) adj (embolism or infarct*)).tw.	57628
36	*multiple trauma/	7535
37	(severe trauma or multiple trauma).tw.	8120
38	*head injury/	23615
39	*brain injury/	49447
40	(traumatic brain injur* or TBI or head trauma or craniocerebral trauma).tw.	83463
41	*cerebrovascular accident/	88315
42	*stroke unit/	1708
43	stroke.tw.	429663
44	*sepsis/	54747
45	*septic shock/	20597
46	(sepsis or septic shock).tw.	187198
47	*brain hemorrhage/	36063
48	((intracranial or intra-cranial or subarachnoid or cerebral or life-threatening) adj2 (hemorrhage or haemorrhage or bleeding)).tw.	71928

49	*intoxication/	29404
50	(severe adj (poisoning or intoxication*)).tw.	1441
51	*diabetic ketoacidosis/	6169
52	diabetic ketoacidosis.tw.	10203
53	*acute liver failure/	3996
54	(acute hepatic failure* or fulminating hepatic failure* or acute liver failure*).tw.	13230
55	*acute kidney failure/	39712
56	(acute kidney failure* or acute renal injur*).tw.	3360
57	*intestine perforation/	4634
58	*appendicitis/	9151
59	(intestinal perforation* or appendicitis).tw.	25969
60	((acute or emergency) adj2 (surgery or surgical or operat* or resection)).tw.	41419
61	*coronavirinae/	1883
62	*coronaviridae infection/	94
63	*coronavirus disease 2019/	145728
64	(coronavirus or corona virus or covid or covid2019 or covid19 or SARS2 or SARS-CoV-2 or SARS-CoV-19 or severe acute respiratory pneumonia outbreak or novel cov or 2019ncov or sars cov2 or cov22 or ncov or coronaviridae).tw.	222621
65	12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64	3029426
66	11 and 65	124288
67	CROSSOVER PROCEDURE.sh.	69047
68	DOUBLE-BLIND PROCEDURE.sh.	191099
69	SINGLE-BLIND PROCEDURE.sh.	44757
70	(crossover* or cross over*).ti,ab.	115800
71	placebo*.ti,ab.	336555
72	((single or double or triple or treble) adj (blind* or mask*)).ti,ab.	258648
73	allocat*.ti,ab.	175818

74	trial.ti.	346935
75	RANDOMIZED CONTROLLED TRIAL.sh.	689904
76	random*.ti,ab.	1739398
77	67 or 68 or 69 or 70 or 71 or 72 or 73 or 74 or 75 or 76	2207781
78	(exp animal/ or exp invertebrate/ or animal.hw. or nonhuman/) not (exp human/ or human cell/ or (human or humans or man or men or wom?n).ti.)	6743066
79	77 not 78	1957342
80	66 and 79	12062
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82	(Relationship between inspired oxygen concentration and atelectasis formation after extubation).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword heading word, floating subheading word, candidate term word]	1
83	81 or 82	9
84	80 or 83	12062

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#1:
TI=((inspir* or inhal* or fraction* or concentrat* or arterial* or saturation or level* or tension* or supply* or supplement* or supplie* or therap* or administr* or dosag* or dose* or dosing*) NEAR/3 oxygen) or TI=(hyperoxia or hyperoxemia or hyperoxaemia or hypoxia or hypoxemia or hypoxaemia or anoxia or anoxemia or anoxaemia or "arterial oxygen" or "high oxygen" or oxygenat* or "blood gas" or "oxygen saturation" or pao2 or sao2 or spo2 or fio2) (249,189 records)

#2:
TS=("emergency department*" or ED or "emergency room*" or ER or "high dependency unit*" or HDU or prehospital* or pre-
hospital* or "critical illness" or "critically ill" or "acutely ill" or "intensive care" or "critical care" or ICU* or "coronary care unit*" or "neurological intermediate care unit*") or TS=("cardiac arrest" or "cardiac failure*" or CPR or "heart arrest" or "heart failure*" or "myocardial infarct*" or "myocardial ischemia" or "acute coronary syndrome") or TS=(shock or meningitis or pneumonia or COPD or "chronic obstructive pulmonary disease" or "acute lung injur*" or ALI) or TS=((adult or acute) near/2 (Respiratory
Distress)) or TS=(((pulmonary or lung) near/2 (embolism or
infarct*)) or "severe trauma" or "multiple trauma" or "traumatic brain injur*" or TBI or "head trauma" or "craniocerebral trauma") or TS=(stroke or sepsis or "septic shock") or TS=((intracranial or intra-
cranial or subarachnoid or cerebral or life-threatening) NEAR/2 (hemorrhage or haemorrhage or
bleeding)) or TS=("severe poisoning" or "servere intoxication'" or "diabetic ketoacidosis" or "acute hepatic failur*" or "fulminating hepatic failur*" or "acute liver failur*") or TS=("acute kidney failur*" or "acute renal injur*" or "intestinal perforation*" or appendicitis or ((acute or emergency) near/2 (surgery or surgical or operat* or
resection*))) or TS=(coronavirus or "corona virus" or covid or covid2019 or covid19 or SARS2 or SARS-CoV-2 or SARS-CoV-
19 or "severe acute respiratory pneumonia outbreak" or "novel cov" or 2019ncov or "sars cov2" or cov22 or ncov or coronaviridae) (5,684,720 records)

#3: 1 and 2 (32,766 records)

#4: TS=((control* OR clinical OR comparative) NEAR/3 (trial* or
stud*)) OR TS=random* OR TS=((single or double or triple or treble) NEAR/6 (mask* or
blind*)) OR TS=(crossover OR cross-
over) OR TS=(factorial* OR placebo* OR volunteer*) OR TI=(trial* or groups) (9,432,324 records)

#5: 3 and 4 (5722 records)

LILACS 06.01.2022 – 643 hits

http://bases.bireme.br/cgi-bin/wxislind.exe/iah/online/

Search history details:

(hyperoxia or hyperoxemia or hyperoxaemia or hypoxia or hypoxemia or hypoxaemia or anoxia or anoxemia or anoxaemia or oxygen or blood gas or pao2 or sao2 or spo2 or fio2) [Words] AND (acute surgery or acute operation or acute resection or emergency surgery or emergency operation or emergency resection) or (intestinal perforation or appendicitis) or (acute kidney failure or acute renal injuries) or (acute hepatic failure or fulminating hepatic failure) or (diabetic ketoacidosis) or (severe poisoning) or (intracranial hemorrhage or subarachnoid hemorrhage or cerebral hemorrhage or intracranial bleeding or life-threatening bleeding) or (sepsis or septic shock) or (stroke) or (traumatic brain injury or TBI or head trauma or craniocerebral trauma) or (severe trauma or multiple trauma) or (pulmonary embolism or pulmonary infarction) or (adult respiratory distress syndrome or ARDS) or (acute lung injury) or (COPD or chronic obstructive pulmonary disease) or (pneumonia) or (meningitis) or (shock) or (cardiac arrest or cardiac failure or CPR or heart arrest or heart failure or myocardial infarction or myocardial ischemia or acute coronary syndrome) or (emergency department or ED or emergency room or ER or high dependency unit or HDU or prehospital or critically ill or acutely ill or intensive care or critical care or ICU or coronary care unit or neurological intermediate care unit or coronavirus or corona virus or covid or SARS2 or SARS-CoV-2 or SARS-CoV-19 or severe acute respiratory pneumonia outbreak or novel cov or 2019ncov or sars cov2 or cov22 or ncov or covid19 or corona viridae) [Words] AND (randomized or randomised or random or randomly OR control or controlled OR RCT OR placebo OR group OR trial) [Words]

Data collection form

General				Patient information + Randomisation and follow-up (L=Lower oxygenation strategy; H=Higher oxygenation strategy)						
Trial ID	Country	Year	Publ.type	L: No. randomised	H: No. randomised	Total randomised	L: No. analysed	H: No. analysed	Total analysed	L: Lost to follow-up

Patient information + Randomisation and follow-up (L=Lower oxygenation strategy; H=Higher oxygenation strategy)											
H: Lost to follow-up	L: Number analysed at longest follow-up	H: Number analysed at longest follow-up	L: Number analysed at secondary time point	H: Number analysed at secondary time point	L: mean or median age	H: mean or median age	L: proportion of male sex	H: proportion of male sex	Medical population (Y/N)	Surgical population (Y/N)	Mixed population (Y/N)

Outcome X (longest follow-up/secondary time-point)									
Assessment (Y/N)	Type of assessment	Mean /median	L: number analysed	L: mean/median	L:SD/IQR	H: number analysed	H: mean/median	H:SD/IQR	Time point used

Characteristics of included trials

Asfar 2017[2]

Study characteristics	
Methods	RCT 2-by-2 factorial trial randomising to 4 groups; 2 groups are described in this review.
Participants	**Sample size:** 442 randomised (experimental 219, control 223) **Sex(male):** experimental 63%, control 66% **Age (mean):** experimental 67.8, control 66.3 Country: France **Setting:** multidisciplinary ICU Disease severity score: SAPS III median 71 Inclusion criteria: 1. Patients aged 18 years and older if they were mechanically ventilated and exhibited septic shock refractory to fluid resuscitation as defined by an absence of response to 20 mL/kg of crystalloids or colloids and requiring vasopressor (norepinephrine or epinephrine, at a minimum infusion rate of 0.1 µg/kg per min); they also had to have been assessed within 6 hours after the initiation of vasopressors. Septic shock was defined by the presence of 2 or more diagnostic criteria of systemic inflammatory response syndrome, proven or suspected infection, and sudden dysfunction of at least 1 organ. Exclusion criteria: 1. Severe hypoxaemia defined as PaO_2: FiO_2 ratio of less than 100 mmHg for a minimum positive end-expiratory pressure of 5 cm H_2O 2. Plasma sodium concentration of less than 130 mmol/L or more than 145 mmol/L 3. Intracranial hypertension 4. Patient admitted for cardiac arrest 5. Overt cardiac failure 6. Under legal guardianship 7. No affiliation with the French healthcare system

	8. Pregnancy 9. Recent participation in another biomedical study or another interventional study with mortality as the primary endpoint 10. An investigator's decision not to resuscitate
Interventions	**Experimental**: hyperoxia group (mechanical ventilation with FiO_2 of 1.0 for 24 hours after inclusion; thereafter FiO_2 as in the normoxia group). **Control**: target SaO_2 of 88% to 95% using mechanical ventilation **Co-intervention**: not specified **Duration**: 24 hours
Outcomes	Primary outcome 1. All-cause mortality at day 28 after inclusion Secondary outcomes 1. 90-day mortality 2. Daily SOFA from inclusion to day 7 3. Days alive and free from organ dysfunction at day 28 4. Length of stay in the ICU 5. Alive at day 28 without organ support was defined as days alive without vasopressor infusion, mechanical ventilation, or renal replacement treatment 6. Safety data (as specified in protocol (NCT01722422)) Outcomes not prespecified 1. Participants with at least 1 serious adverse event 2. Chest radiograph scores 3. Atelectasis 4. Pneumothorax 5. Ventricular arrhythmias 6. Mesenteric ischaemia 7. Digital ischaemia 8. ICU-acquired weakness 9. Participants with ≥1 nosocomial infection during ICU stay

	10.Participants with ≥1 nosocomial pneumonia during ICU stay
Notes	Prospectively registered (NCT01722422)

Barrot 2020[3]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 205 patients randomised out of 850 patients planned (experimental 102, control 103) **Sex (male):** Experimental 62.7%, Control 65.7% **Age (mean):** Experimental 63.5, Control 63.0 Country: France **Setting:** Mixed disciplinary intensive care units **Disease severity score:** SOFA score median 9, SAPS III score median 67 Inclusion criteria 1. Adult patients receiving invasive mechanical ventilation for less than 12 hours for ARDS (according to Berlin Criteria) Exclusion criteria 1. Pregnancy 2. Children (age < 18 years) 3. Incapacitated patients or under legal guardianship 4. Home oxygen therapy 5. Non-invasive ventilation at home (except due to sleep apnea syndrome) 6. Resuscitated from cardiac arrest 7. Traumatic brain injury or intra cranial hypertension 8. Hemoptysis requiring embolisation or surgery 9. Patient with imminent risk of death within 24 hours or dying patient 10.Patient receiving extra corporal membrane oxygenation or extra corporal life support at admission 11.Chronic obstructive pulmonary disease 12.Use of hyperbaric oxygen therapy

	13.Cyanide poisoning or methemoglobinemia 14.Non-drained pneumothorax 15.Lymphangitic carcinomatosis 16.Eosinophilic lung disease 17.Resuscitated for organ donation 18.Inclusion in other trials with_where_mortality being_is_the primary outcome
Interventions	**Experimental:** PaO_2 90-105 mmHg; SpO_2 ≥96%. **Control:** PaO_2 55-70 mmHg; SpO_2 88-92%. **Co-intervention:** Ventilation and weaning protocol (as stated in the protocol) **Duration:** 7 days or until extubation (if before 7 days)
Outcomes	Primary outcome 1. All-cause mortality at day 28 after randomisation Secondary outcomes 1. ICU mortality 2. 90-day mortality 3. Sequential Organ Failure Score (without respiratory component) at days 0, 3 and 7 4. Days alive without mechanical ventilation within 28 days from randomisation 5. Ventilator associated pneumonia during the first 28 days 6. Septicaemia during the first 28 days 7. Consumption of antibiotics (days exposed/days in the intensive care unit) 8. Cardiovascular complications: new-onset arrhythmia, cardiac ischemic event or use of vasopressors over the first 7 days 9. Successful weaning from mechanical ventilation at days 28 and 90 10.Neurologic status according to the daily Richmond Agitation and Sedation Scale 11.Seizures 12.New cerebral stroke 13.Daily sedation doses

	14.Administration_of neuroleptics Not prespecified 1. Delirium 2. Mesenteric ischemia
Notes	Prospectively registrered (NCT02713451).

Gelissen 2021[4]

Study characteristics	
Methods	RCT
Participants	**Sample size**: 574 (experimental 280, control 294) **Sex (male %)**: experimental 65%, control 65% **Age (median)**: experimental 68, control 68 **Country**: The Netherlands **Setting**: multidisciplinary ICU **Disease severity score**: Admission SOFA score (excluding the respiratory component) 6 in experimental, 5 in control Inclusion criteria 1. ≥18 years 2. ≥2 positive SIRS-criteria: a. Temperature >38 °C or hypothermia (<36 °C) b. Heart rate >90 beats per minute c. Respiratory rate >20 /min or pCO2 <32 mmHg (4.3 kPa) d. Number of leucocytes >12 x 10^9/l or <4 x 10^9/l or >10% bands 3. Within 12 hours of admittance to the ICU 4. Expected stay of `48 hours as estimated by the attending physician Exclusion criteria 1. Elective surgery 2. Carbon monoxide poisoning 3. Cyanide intoxication 4. Methemoglobinemia

	5. Sickle cell anaemia 6. Severe pulmonary arterial hypertension (WHO class III or IV) 7. Known severe Acute Respiratory Distress Syndrome (ARDS) (PaO2/FiO2 ≤100 mmHg and PEEP ≥5 cm H_2O) 8. Known cardiac right to left shunting 9. Pregnancy 10.Severe Chronic Obstructive Pulmonary Disease (COPD) (Gold class III or IV) or other severe chronic pulmonary disease 11.Patients participating in other interventional trials
Interventions	**Experimental**: oxygen therapy was administered to target a PaO_2 range of 14 to 18 kPa. The target PaO_2 was achieved by adjustment of the FIO_2 and/or PEEP. Levels to achieve the target PaO_2 were maximized to a FIO2 of 0.60 and 10 cm H_2O of PEEP unless clinically otherwise indicated. **Control**: oxygen therapy was administered to target a PaO_2 range of 8 to 12 kPa. The target PaO_2 was achieved by adjustment of the FIO_2 and/or PEEP. Levels to achieve the target PaO_2 were maximised to an FIO_2 of 0.60 and 10 cm H2O of PEEP unless clinically otherwise indicated. Co-intervention: not specified **Duration:** 14 days or until ICU discharge or death.
Outcomes	Primary outcome 1. Daily Delta Sequential Organ Failure Assessment Score within 14 days (excluding respiratory system score) Secondary outcomes 1. Total maximum SOFA score minus SOFA score on admission within 14 days (excluding respiratory system score) 2. SOFA rate of decline within 14 days (excluding respiratory system score) 3. Total maximum SOFA score (excluding respiratory system score) 4. ICU mortality 5. In-hospital mortality 6. Hypoxic events (PaO2 <55 mmHg) within 14 days 7. Vasopressor / Inotrope requirements within 14 days 8. Renal function, fluid balance within 14 days 9. Duration of mechanical ventilation within 14 days

	10.Ventilator-free days within 14 days 11.Length of ICU-stay Secondary outcomes not prespecified 1. Intubation during ICU-admission 2. 90-day mortality 3. Number of hyperoxic measurements 4. Proportion of patients with one or more serious adverse events (including death) 5. New myocardial infarction 6. New liver failure 7. New Stroke 8. Severe respiratory failure necessitation prone ventilation Outcomes not reported on: 1. Time spent in assigned PaO2 range 2. Mortality within 14 days 3. Oxidative stress (F2-isoprostanes) within 14 days 4. Systemic Vascular Resistance Index in a random subpopulation, within 14 days 5. Cardiac Index in a random subpopulation, within 14 days 6. Microcirculatory flow index and perfused vessel density in a random subpopulation, within 14 days 7. Length of hospital-stay
Notes	Prospectively registered (NCT02321072).

Girardis 2016[5]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 480 (experimental 244, control 236) **Sex (male %):** experimental 57%, control 56% **Age (median):** experimental 65, control 63 Country: Italy **Setting:** multidisciplinary ICU

	Disease severity score: SAPS II score median 38 Inclusion criteria 1. All patients aged 18 years or older and admitted to the ICU with an expected length of stay of 72 hours or longer Exclusion criteria 1. Age younger than 18 years 2. Pregnancy 3. ICU readmission 4. A decision to withhold life-sustaining treatment 5. Immunosuppression or neutropenia 6. Enrolment in another study 7. Patients with acute decompensation of COPD and ARDS with a PaO_2:FiO_2 ratio <150 mmHg
Interventions	**Experimental:** oxygen therapy was administered according to standard ICU practice; FiO_2 of at least 0.4, allowing PaO_2 values up to 150 mmHg and an SpO_2 between 97% and 100%. If the SpO2 decreased below 95% to 97%, the FiO_2 was increased to reach the target value of SpO_2 . Participants received FiO2 of 1.0 during intubation, airway suction, or hospital transfer. **Control:** oxygen therapy was administered at the lowest possible FiO_2 to maintain the PaO_2 between 70 and 100 mmHg or SpO_2 values between 94% and 98%. FiO_2 was gradually reduced, or oxygen supplementation discontinued whenever the PaO_2 or SpO_2 exceeded 100 mmHg or 98%. Supplemental oxygen was administered only if SpO_2 decreased below 94%. Co-intervention: not specified **Duration:** until ICU discharge
Outcomes	1. ICU mortality 2. New-onset respiratory, cardiovascular, liver, and renal failure (defined as a SOFA score ≥ 3 for the corrresponding organ) occurring 48 hours or more after ICU admission 3. Need for reoperation in surgical patients

	4. Bloodstream, respiratory, and surgical site infections (according to Centers for Disease Control and Prevention definitions). Only microbiologically documented bloodstream and respiratory tract infections were considered. Secondary outcomes not prespecified 1. Hospital mortality 2. Ventilation-free hours during the ICU stay
Notes	Trial registration: NCT01319643.

Gomersall 2002[6]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 36 (experimental 19, control 17) **Sex (male %):** experimental 82%, control 76% **Age (mean):** experimental 68, control 69 **Country:** Hong Kong **Setting:** multidisciplinary ICU Disease severity score: not reported Inclusion criteria 1. Patients admitted with a clinical diagnosis of an acute exacerbation of COPD and a PaO2 < 6.6 kPa (50 mmHg), and PaCO2 > 6.6 kPa (50 mmHg) on air. Exclusion criteria 1. Chest radiologic signs of pulmonary oedema, lung cancer, pneumothorax, or pneumonia 2. If the patient already met study criteria for mechanical ventilation 3. Mechanical ventilation for respiratory failure twice in the preceding 6 months 4. Inability to walk more than 20 yards on flat ground 5. Co-existing terminal disease

Interventions	Oxygen therapy was provided via a Venturi-type mask and adjusted according to the results of arterial blood samples with the aim of reaching the desired target oxygen tension within 1 hour of trial entry. **Experimental:** target PaO_2 above 9.0 kPa (70 mmHg **Control:** target PaO_2 of > 6.6 kPa (50 mmHg) **Co-intervention**: participants in the low-oxygen tension group also received doxapram if they developed an acidosis with pH < 7.2, whereas those in the high-oxygen tension group received doxapram if they developed symptomatic acidosis. Bronchodilator, steroid, and antibiotic therapy was standardized. **Duration:** treatment protocols, including oxygen therapy, were continued after discharge from the ICU until oxygen therapy was no longer considered necessary
Outcomes	1. Need for mechanical ventilation 2. Duration of hospital stay 3. Cardiac arrhythmia 4. Mortality 5. Coma 6. Lung function tests at discharge and at 6 months
Notes	Email sent to Dr Gomersall 29 November 2021, but no reply was received

Ishii 2018[7]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 44 (experimental 21, control 23) **Sex:** not specified **Age:** not specified Country: Japan **Setting:** surgical ICU Disease severity score: not reported **Inclusion criteria:** mechanically ventilated patients admitted to surgical ICU for more than 12 hours Exclusion criteria: not specified
Interventions	**Experimental:** FiO_2 of 1.0 using high-flow nasal cannula.

	Control: expected FiO_2 to achieve a PaO_2 of 100 mmHg (13.3 kPa) using high-flow nasal cannula The interventions are 'non-invasive', as they are initiated after extubation (of the mechanical ventilated), whereas after oxygen they are administered via high-flow nasal cannula. Co-intervention: not specified Duration: 1 hour
Outcomes	1. Atelectasis
Notes	Full text was not found.

Jakkula 2018[8]

Study characteristics	
Methods	RCT with a 2-by-3 factorial design. We only extracted data from the normoxia and moderate-hyperoxia groups.
Participants	**Sample size:** 123 (experimental 60, control 63) **Sex (male %):** experimental 48%, control 50% Age: experimental 60, control 59 **Country:** Finland and Denmark **Setting:** adults admitted to the ICU after OHCA **Disease severity score:** APACHE II score median 28 Inclusion criteria 1. Adults resuscitated from witnessed OHCA with VF or VT as the initial rhythm. In addition, all of the following inclusion criteria had to be met: a. ROSC 10 to 45 minutes from the onset of cardiac arrest; b. confirmed or suspected cardiac origin of the arrest; c. mechanical ventilation upon ICU arrival; d. markedly impaired level of consciousness defined as no response to verbal commands and GCS motor score < 5 (withdrawal to painful stimuli at best); e. deferred consent from next of kin possible or likely; and f. active intensive care and TTM initiated. Exclusion criteria 1. Adults with confirmed or suspected acute or pre-existing intracranial pathology or suspicion of increased intracranial pressure, or both

	2. Adults with severe oxygenation failure defined as PaO_2 /FiO_2 < 100 mmHg upon arrival to ICU and no improvement in oxygenation after adding sufficient PEEP level 3. Severe COPD 4. Age < 18 or > 80 years 5. Pregnancy
Interventions	**Experimental:** target PaO_2 of 20 to 25 kPa (150 to 187.5 mmHg). **Control:** target PaO_2 of 10 to 15 kPa (75 to 112.5 mmHg) or target SpO2 of 95% to 98%. **Co-intervention:** all adults received TTM at 33 °C or 36 °C and were sedated according to the treating clinicians' instructions. All adults received standard care, monitoring and assessments based on the protocol of the ICU, including direct blood pressure monitoring via an arterial catheter. **Duration:** 36 hours
Outcomes	Primary outcome 1. NSE serum concentration at 48 hours after cardiac arrest Secondary outcomes 1. NSE serum concentration at 24 and 72 hours after cardiac arrest 2. S100 protein serum concentration at 24, 48, and 72 hours after cardiac arrest 3. Troponin-T concentration at 24, 48, and 72 hours after cardiac arrest 4. Results of NIRS monitoring during the first 48 hours after admission to the ICU 5. Results of continuous EEG monitoring for 48 hours after arrival at the ICU and a statement of the findings by an experienced senior neurologist or neurophysiologist 6. CPC at 6 months after cardiac arrest 7. Total duration of intensive care 8. Total duration of mechanical ventilation 9. Length of hospital stay 10.Discharge destination 11.Vital status at hospital discharge (dead or alive) Feasibility outcomes 1. Difference in $PaCO_2$ between groups targeting low to normal (4.5 to 4.7 kPa) and high to normal (5.8 to 6.0 kPa) $PaCO_2$

	2. Difference in PaO_2 between groups targeting low to normal (10 to 15 kPa) and high to normal (20 to 25 kPa) PaO_2 3. Difference in MAP between groups targeting low to normal (65 to 75 mmHg) and high to normal (80 to 100 mmHg) MAP 4. Distribution of values for primary and secondary outcomes 5. Randomized or screened participant ratio 6. Consent rate 7. Data completion rate 8. Recruitment duration
Notes	Prospectively registered (NCT02698917)

Jun 2019[9]

Study characteristics	
Methods	RCT Three factor trial design. Only included the moderate concentration oxygen group and high concentration oxygen
Participants	**Sample size:** 87 (allocation not specified) **Sex (male):** not specified **Age (median):** not specified **Country:** not specified **Setting:** not specified Disease severity score: not reported Inclusion criteria 1. Elderly patients with acute exacerbation of COPD and AMI not accepting percutaneous coronary intervention. Exclusion criteria 1. Shock 2. Malignant arrythmia

Interventions	**Experimental:** High concentration oxygen group: Invasive ventilation with PRVC mode and a two-step FiO_2 -management: 1) FiO2 50-70% within 48 hours 2) FiO_2 from 50% to 40% gradually over 48 hours. **Control:** Moderate concentration oxygen group: Invasive ventilation with PRVC mode and FiO_2 30-50%. **Duration:** 96 hours
Outcomes	1. 14-day mortality 2. Occurrence of malignant arrhythmia 3. Extubation-rate 4. Recurrence rate of myocardial infarction 5. Occurrence of hyperoxia
Notes	Full text was not found.

Lång 2018[10]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 65 (experimental 38, control 27) **Sex (male, %):** experimental 82%, control 85% **Age:** experimental 45, control 43 **Country:** Finland **Setting:** mechanically ventilated adults with traumatic brain disease admitted to the ICU **Disease severity score:** APACHE II score median 22 Inclusion criteria 1. Isolated non-penetrating TBI or adults with multiple trauma with TBI with GCS 8 or less (inclusive), expected need for intubation and mechanical ventilation > 24 hours 2. Recruitment within 18 hours after admission to ICU 3. Time from TBI < 36 hours 4. Informed consent from next of kin Exclusion criteria

	1. Age < 18 or > 65 years 2. Anticipated brain death in 12 hours or otherwise moribund adults expected to die in 24 hours 3. Expected need for mechanical ventilation < 24 hours 4. Insufficient oxygenation assessed by a clinician 5. Adults with multiple trauma with brain injury and severe abdominal, thoracic, or pelvic injury possibly affecting oxygenation 6. No consent 7. Insufficient oxygenation with the treatment modality of the lower oxygenation group (PaO2 < 13 kPa or SpO2 < 95% with FiO2 of 0.40 and PEEP of 10) 8. Oxygenation failure probable during ICU care 9. Penetrating TBI 10.Suspected pregnancy (perform urinary or serological pregnancy test if suspected)
Interventions	**Experimental:** FiO_2 of 0.70. **Control:** FiO_2 of 0.40. Co-intervention: not specified **Duration:** maximum 14 days
Outcomes	1. Laboratory markers during the first 3 days 2. Pulmonary function (PaO2:FiO2 ratio, ARDS, atelectasis, pneumonia) 3. Length of mechanical ventilation 4. Length of ICU stay 5. Length of hospital stay 6. Death at 6 months 7. Extended Glasgow Outcome Scale
Notes	Email sent 25 January 2022 to Dr. Lång and a reply was received. Prospectively registered (NCT01201291)

Mackle 2020[11]

Study characteristics	
Methods	RCT

Participants	**Sample size:** 1000 (experimental: 501, control: 499) **Sex (male, %):** Experimental 62.8%, Control 63.2% **Age (mean):** Experimental 58.1, Control 57.5 **Country:** Australia and New Zealand **Setting:** Mechanically ventilated adults admitted to multidisciplinary ICU **Disease severity score:** APACHE II mean score; Experimental 23.6, Control 23.3 Inclusion criteria 1. Patients ≥18 years who were expected to receive mechanical ventilation in the ICU beyond the next calendar day after recruitment Exclusion criteria 1. Greater than 2 hours of either invasive mechanical ventilation, or non-invasive ventilation, or both, in an ICU during this hospital admission (including time ventilated in another hospital's ICU) 2. In the view of the treating clinician, hyperoxia is clinically indicated for reasons including (but not limited to) carbon monoxide poisoning or a requirement for hyperbaric oxygen therapy 3. In the view of the treating clinician, avoidance of hyperoxia is clinically indicated for reasons including (but not limited to) chronic obstructive pulmonary disease (added after enrolment of first 100 patients), paraquat poisoning, previous exposure to bleomycin, or chronic hypercapnic respiratory failure 4. Pregnancy 5. Death is deemed to be inevitable as a result of the current acute illness and either the treating clinician, the patient or the substitute decision maker are not committed to full active treatment 6. Patients with a life expectancy < 90 days due to a chronic or underlying medical condition 7. Patients admitted after a drug overdose (including alcohol intoxication) 8. Long-term dependence on invasive ventilation prior to this acute illness 9. Confirmed or suspected diagnosis of any of the following: Guillain–Barré syndrome, cervical cord injury above C5, muscular dystrophy, or motor neurone disease 10.Enrolment not considered in the patient's best interests 11.Enrolled in any other trial of targeted oxygen therapy (added after enrolment of first 100 patients)

	12.Previously enrolled in the ICU-ROX study
Interventions	**Experimental:** SpO2 90-100%, FiO_2<0.30 discouraged. Patients assigned to the ´higher group´ received ´standard of care´ during mechanical ventilation and after extubation. No specific measures were taken to avoid a high FiO_2 or a high SpO_2. The use of upper alarm limits for the SpO_2 in the higher group was prohibited, as upper alarm limits for the SpO_2 were not used as part of standard of care. The lower limit alarm for the SpO_2 were not used as part of standard of care. The lower limit alarm for the $SpO_2$was set at 90% (or lower if clinically appropriate). If the PaO_2 or the SaO_2 were lower than the acceptable limits, the FiO_2 could be increased irrespective of the SpO_2, if clinically appropriate. A FiO_2 of less than 0.3 during mechanical ventilation was discouraged. **Control:** SpO_2 90-96%. For patients assigned to the 'lower group´, the FiO_2 was decreased to room air (0.21) as rapidly as possible provided that the SpO_2 remained above the acceptable lower limit of 90%. Supplemental oxygen for SpO_2 levels greater than 96% were strictly avoided and an alarm was set to sound at SpO_2 levels above 96% whenever oxygen was administered. If the SpO_2 exceeded the acceptable upper limit, the dosage of supplemental oxygen was gradually decreased as a high priority. The lower limit alarm for SpO_2 was set at 90% (or lower if clinically appropriate). If the PaO_2 or SaO_2 was lower than the acceptable limit, the FiO_2 could be increased irrespective of the SpO2 , if clinically appropriate. **Co-intervention:** there were no restrictions on concomitant treatments provided to patients. A temporary increase in FiO_2 due to procedures performed in the ICU was permitted in both groups. These procedures included but were not limited to: bronchoscopy, suctioning, tracheostomy and preparation for extubation. **Duration:** until death or discharge from the ICU, or day 28 post-randomisation
Outcomes	Primary outcome 1. Number of ventilator-free days from randomisation to day 28 Secondary outcomes 1. All-cause mortality at day 90 and 180 2. Duration of survival 3. Proportion of participants in paid employment at baseline who are unemployed at day 180 4. Cognitive function assessed at day 180 using a telephone interview 5. Health-related quality of life at day 180 6. Functional outcome at day 180 by the Extended Glasgow Outcome Scale
Notes	Email sent 14 January 2022 to Dr. Young and a reply was received.

	Prospectively registered (ACTRN12615000957594).

Martin 2021[12]

Study characteristics	
Methods	RCT
Participants	**Sample size:** Planned 60, included 34 (experimental 17, control 17) **Sex (male, %):** 64.7% overall, not specified for individual groups **Age (median):** 66 overall, not specified for individual groups **Country:** England **Setting:** Mechanically ventilated patients at multidisciplinary ICU **Disease severity score:** SOFA score median 11, APACHE II score median 23 Inclusion criteria 1. Mechanically ventilated adults (18 years of age or older) within 24 hours of an unplanned admission to ICU, who had a diagnosis of respiratory failure and where it was thought mechanical ventilation was expected to be required for 72 hours or more. Exclusion criteria 1. ICU admission following surgery (elective or unplanned) 2. Patients expected to die within 24 hours of admission to ICU 3. Pregnancy 4. Admission post cardiac arrest 5. Chronic lung disease known (or highly suspected) having baseline SpO_2 in the range of the intervention arm (i.e., 88%–92%) 6. Admission post-trauma (including traumatic brain injury) 7. Known sickle cell trait or disease 8. Ongoing significant haemorrhage or profound anaemia 9. Severe peripheral vascular disease 10.Severe pulmonary hypertension 11.Other medical conditions where mild hypoxaemia would be contraindicated 12.Patients participating in other interventional clinical trials
Interventions	**Experimental:** $SpO_2 \geq 96\%$. If a participant was briefly transferred out of ICU (e.g. investigation, or imaging) the trial was paused until they returned to ICU. In the

	protocol this group is characterised as "normal SpO_2 target group" (control in publication). No targets or limitations for PaO_2. **Control:** SpO_2 88-92%. In the protocol this group is characterised as "lower than normal SpO_2 target group" (conservative in publication). If a participant was briefly transferred out of ICU (e.g. investigation, or imaging) the trial was paused until they returned to ICU. No targets or limitations for PaO_2. **Co-intervention:** not specified. **Duration:** until extubation, formation of tracheostomy, transfer to another ICU or death.
Outcomes	Primary outcome 1. Feasibility assessed as a) recruitment rate, b) protocol deviations, c)rate of withdrawal, and d) reasons for any withdrawal Secondary outcomes 1. Oxidative stress 2. Adverse effects 3. Length of critical care stay at 30 and 90 days 4. Length of hospital stay at 30 and 90 days 5. Survival at critical care unit discharge at 30 and 90 days 6. Adverse events Treatment end-points as defined in the study protocol 1. Extubation 2. Formation of tracheostomy 3. Transfer to another critical care unit 4. Death Secondary outcomes not prespecified 1. Support for the trial from clinicias 2. Patient agreement to participate 3. Change in SOFA score
Notes	Terminated prematurely due to end of extended recruitment window. Prospectively registered (NCT03287466).

Mazdeh 2015[13]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 51 (experimental 26, control 25) **Sex (male %):** experimental 54%, control 56% **Age:** not specified Country: Iran **Setting:** adults with stroke initially referred to the Department of Neurology, but admitted to the ICU Disease severity score: Bartel index mean 42 Inclusion criteria 1. Age between 40 and 70 years 2. GCS > 12 and adults with isolated brain damage and intact airway control 3. Ischaemic and haemorrhagic stroke with no need for surgical intervention 4. Less than 12 hours have passed since the accident 5. NIHSS square between 7 and 9 Quote: "Patients were selected from patients referred to the Department of Neurology of Farshchian Hospital, an affiliated hospital of Hamadan University of Medical Sciences. The patients were admitted to the ICU and monitored carefully by expert nurses." Due to participants being transferred from the Department of Neurology to the ICU to be monitored, we do not regard these adults as typical adults admitted to the ICU. Exclusion criteria 1. Adults under 40 and older than 70 years 2. Adults with diabetes mellitus and ischaemic heart disease, renal failure, acute pulmonary oedema, history of massive myocardial infarction, and heart failure 3. Adults who need intubation on arrival in the hospital 4. Adults with a baseline blood pressure of less than 90/60, or hypoxia 5. Adults requiring surgical intervention (i.e. acute subdural haematoma and cerebral haemorrhage) 6. Adults with blood pressure greater than 170/90 in the first 12 hours of the incident

	7. Adults with successful CPR within 12 hours 8. History of previous stroke or unconsciousness resulting in the need for intubation and mechanical ventilation 9. Death or lost to follow-up 10.Adults in the control group for whom oxygen therapy was inevitable
Interventions	**Experimental:** FiO_2 of 0.5 - oxygen therapy with Venturi mask **Control:** no supplemental oxygen was administered **Co-intervention:** routine medication (as stated in protocol) **Duration:** oxygen administration was given in the first 12 hours of admission
Outcomes	1. Good recovery and lower number of complications in the first day of admission, before discharge, and 6 months after discharge using the modified Rankin scale and Barthel Index (as stated in protocol) 2. Outcome not prespecified: mortality
Notes	Retrospectively registered (IRCT201212199647N2)

Panwar 2016[14]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 104 (experimental 51, control 53 (1 lost to follow-up)) **Sex (male %):** experimental 65%, control 62% **Age:** experimental 62, control 62 **Country:** Australia, New Zealand, and France **Setting:** mechanically ventilated adults admitted to a multidisciplinary ICU **Disease severity score:** APACHE III score median 80 (control) and 70 (experimental) Inclusion criteria 1. People admitted to the ICU 2. Aged ≥ 18 years 3. Receiving invasive MV for < 24 hours, and their treating clinician expected MV to continue for at least the next 24 hours The reason for the inclusion criterion of receiving invasive MV for < 24 hours was to ensure that participants who would be assigned to the conservative oxygen

	group were not exposed to standard liberal oxygen therapy for prolonged periods prior to randomization. Exclusion criteria 1. Known pregnancy 2. Imminent risk of death 3. If the treating clinician lacked equipoise for the patient to be enrolled in this trial The exclusion criterion "lacked equipoise" included those clinical situations where the most appropriate approach (conservative versus liberal) to oxygen therapy is well established. For example, in hypercapnic patients with chronic respiratory failure or exacerbation of COPD, there is level I evidence supporting a conservative approach to oxygen therapy (1), and in patients with carbon monoxide poisoning or necrotizing fasciitis a liberal approach is preferred. However, amongst patients who had COPD listed as 1 of the prior comorbid conditions, the treating clinicians could permit enrolment of those adults who were admitted for reasons unrelated to COPD.
Interventions	**Experimental:** SpO_2 target ≥ 96%. **Control:** target SpO_2 of 88% to 92%. When FiO_2 requirement was < 0.50, a SpO_2 of 90% to 92% was recommended, and when FiO_2 requirement was ≥ 0.50, a SpO_2 of 88% to 90% was recommended. **Co-intervention:** participating sites were requested to adhere to best practice guidelines in relation to other potentially confounding co-interventions such as adjustment of tidal volume, PEEP, fluid management, blood transfusion, muscle relaxation, sedation interruption, ventilator weaning, nutrition, use of steroids, early mobilization, and physiotherapy. **Duration:** entire duration of mechanical ventilation
Outcomes	Primary outcomes 1. Proportion of time spent in the assigned SpO_2 range in each arm 2. Area under the curve for PaO_2, FiO_2, and SpO_2 on day 0 to day 7 in each arm Secondary outcomes 1. Incidence of circulation-related events 2. Incidence of respiration-related events 3. Incidence of acute kidney injury 4. Incidence of outcomes related to other organ systems 5. Time to successful extubation (alive and extubated for > 48 hours)

	6. MV-free days 7. ICU mortality 8. Hospital mortality 9. 90-day mortality
Notes	Prospectively registered (ACTRN12613000505707).

Schjørring 2021[15]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 2928 (experimental 1466, control 1462) **Sex (male):** experimental 64.9%, control 63.7% **Age (median):** experimental 70, control 70 Country: Denmark, Switzerland, Finland, the Netherlands, Norway, the United Kingdom, and Iceland **Setting:** Acute hypoxaemic respiratory failure at multidisciplinary ICUs Disease severity score: SOFA score median 8 Inclusion criteria 1. Adults (≥ 18 years), receiving ≥ 10 litres of oxygen per minute in an open system or an FiO_2 of ≥0.50 in a closed system, acutely admitted to the ICU, expected to received oxygen supplementation in the ICU for at least 24 hours, and having an arterial cannula in place. Exclusion criteria 1. Cannot be randomized within 12 hours of ICU admission: defined as 12 full hours from the time of the present ICU admission; if the patient is transferred from another ICU, the 12 hours will count from the time of the admission to the first ICU 2. Receives chronic mechanical ventilation (invasive mechanical ventilation, continuous non- invasive ventilation or continuous mask-CPAP) for any reason. Nocturnal CPAP or non-invasive ventilation due to sleep apnea and/or obesity hypoventilation syndrome is not regarded as chronic mechanical ventilation 3. Use of supplementary oxygen at home (supplementary oxygen given through nasal cannula, mask or tracheostomy on a regular daily basis independent of whether it is continuous, in daytime or nocturnal) 4. Previous treatment with bleomycin (any history of bleomycin treatment documented in the patient charts)

	5. Organ transplant (any kind of solid organ transplant planned or performed during current hospitalization) 6. Withdrawal from active therapy or brain death deemed imminent (clinicians or investigators judge that withdrawal from active therapy or brain death is likely within a few hours) 7. Pregnancy confirmed by a positive urine or plasma human chorionic gonadotropin (hCG) (in women 3% for non-smokers or >10% for active smokers during current hospitalization) 9. Cyanide poisoning (if suspected by the clinicians during current hospitalization and documented in the patient charts) 10.Paraquat poisoning (if suspected by the clinicians during current hospitalization and documented in the patient charts) 11.Methemoglobinemia (a confirmed arterial or venous blood methemoglobin >8% during current hospitalization) 12.Sickle cell disease (any history of sickle cell disorder documented in patient charts or the presence of hemoglobin S (HbS) in an arterial or a venous blood sample) 13.Any condition expected to involve the use of hyperbaric oxygen (HBO) treatment (e.g. necrotizing soN tissue infection in sites using HBO for this condition) 14.Consent not obtainable according to national regulations (patients where the clinician or investigator is unable to obtain the necessary consent before inclusion of the patient according to the national regulations) 15.Previously randomized into the HOT-ICU trial (previously screened patients may be eligible if they were not randomized
Interventions	**Experimental:** PaO_2 of 12 kPa (90 mmHg) **Control:** PaO_2 8 kPa (60 mmHg) Co-intervention: not specified **Duration:** until discharge from ICU, death. Including ICU readmission. Maximum 90 days
Outcomes	Primary outcome 1. All-cause mortality at 90 days Secondary outcomes 1. Proportion of patients with one or more of the following serious adverse events (SAEs) the ICU after randomization (new myocardial ischemia, new ischemic stroke, new intestinal ischemia, or new episode of shock)

	2. Percentage of days alive without the use of life-support (mechanical ventilation, renal replacement therapy or circulatory support) within 90 days following randomization 3. Percentage of days alive and out of hospital in the 90-day period
Notes	Prospectively registered (NCT03174002).

Taher 2016[16]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 68 (experimental 34, control 34) **Sex (male %):** experimental 74%, control 68% **Age:** experimental 40, control 46 Country: Iran **Setting:** adults with traumatic brain injury initially referred to the emergency department, but who were admitted to the ICU Disease severity score: GCS score mean 7.4 Inclusion criteria 1. Age between 18 and 65 years 2. Less than 6 hours passed since the accident; haemodynamic stability; and GCS between 3 and 8 Exclusion criteria 1. Pregnancy 2. People under 18 or older than 65 years 3. GCS under 3 or more than 8 4. People with chronic disease such as diabetes mellitus, ischaemic heart disease, renal failure, acute pulmonary oedema, history of massive myocardial infarction, and heart failure 5. People with a baseline blood pressure of less than 90/60 6. People with successful CPR 7. Death or loss to follow-up Participants in the control group for whom oxygen therapy was inevitable were also excluded from this study

Interventions	**Experimental:** FiO_2 of 80% oxygen by mechanical ventilator in the first 6 hours after the traumatic accident. **Control:** FiO_2 of 50% using mechanical ventilator in the first 6 hours after the traumatic accident. Co-intervention: not specified Duration: 6 hours
Outcomes	1. Glasgow Outcome Scale 2. Barthel Index 3. modified Rankin Scale at the time of discharge from hospital and at 6-month follow-up
Notes	Retrospectively registered (IRCT201303209014N17).

Yang 2019[17]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 214 (experimental 114, control 100) **Sex (male, %):** experimental 59.6%, control 69% **Age (median):** experimental 60, control 56 Country: China **Setting:** Adult ICU patients in multidisciplinary ICU **Disease severity score:** APACHE II score median 17 Inclusion criteria 1. Adult (≥ 18 years) ICU patients with an expected ICU stay of at least 72 hours Exclusion criteria 1. ICU readmission 2. Acute exacerbation of chronic obstruction disease 3. Severe acute respiratory distress syndrome (defined as PaO_2:FiO_2 ≥ 100 mmHg) 4. Inclusion in another interventional trial 5. A decision to withhold life-sustaining treatment 6. Pregnancy 7. Paraquat poisoning

	8. More than 12 hours of ICU admission
Interventions	**Experimental:** SpO_2 96-100%, with FiO_2 no lower than 30%. If SpO_2 was unattainable, SaO_2 was used. During intubation, airway suction, tracheostomy and other critical situations such as hemodynamic collapse, physicians could temporarily elevate FiO_2 to 100% **Control:** SpO_2 90-95%, with the FIO_2 as low as possible. If SpO_2 was unattainable, SaO_2 was used. During intubation, airway suction, tracheostomy and other critical situations such as hemodynamic collapse, physicians could temporarily elevate FiO_2 to 100%. Co-intervention: not specified **Duration:** not specified
Outcomes	Primary outcome 1. Mortality at day 28 Secondary outcomes 1. ICU mortality. If patients checked out of ICU in agonal state and died within 24 hours they counted as dead in the ICU. Not predefined secondary outcomes 1. Time-weighted SpO2 2. Time-weighted FiO2 3. Time-weighted PaO2 4. Time to death within 28 days Secondary outcomes not reported 1. Ventilator free time 2. Renal replacement free time
Notes	Prospectively registered (NCT02999932).

Yang 2021[18]

Study characteristics	
Methods	RCT
Participants	**Sample size:** 106 (experimental 55, control 51) **Sex (male):** experimental 63.6%, control 62.7%

	Age (mean): experimental 58.8, control 61.8 Country: China **Setting:** Mechanically ventilated adults with severe pneumonia admitted to the ICU **Disease severity score (mean):** APACHE II score mean 22.2 experimental, 24.17 control Inclusion criteria Patients with severe pneumonia and requiring mechanical ventilation were selected. A combination of 1 major criteria or ≥3 minor criteria according to Infectious Diseases Society of America/American Thoracic Society Consensus Guidelines on the Management of Community-Acquired Pneumonia in Adults were needed(ref). Major criteria: 1. endotracheal intubation and mechanical ventilation 2. septic shock requiring vasopressors Minor criteria: 1. respiratory rate ≥30 breath/min 2. PaO_2:FiO_2 ratio <250 mmHg 3. multilobar infiltrates 4. confusion 5. blood urea nitrogen level, >7.14mmol/L 6. leukopenia resulting from infection (white blood cell count <$4x10^9$/L) 7. thrombocytopenia (platelet count <$100x10^9$) 8. hypothermia (body temperature <36°C) 9. hypotension requiring aggressive fluid resuscitation Exclusion criteria 1. Age<18 years old 2. Pregnancy 3. Withdrawal of any active therapy 4. Immunosuppression or neutropenia 5. Exacerbation of chronic obstructive pulmonary disease

	6. acute respiratory distress syndrome, PaO_2:FiO_2 ratio <150 mmHg
Interventions	**Experimental:** oxygen therapy was administered to target a PaO_2 range of 70-100 mmHg or SpO_2 of 90-92%. **Control:** oxygen therapy was administered to target a PaO_2 >150 mmHg or SpO_2 >96%. Co-intervention: not specified **Duration:** not specified
Outcomes	1. Comparison of blood gas analyses (pH, pCO_2, pO_2) at baseline and after treatment? 2. Mechanical ventilation time 3. ICU length of stay 4. ICU mortality 5. New liver failure 6. New shock 7. New renal failure 8. ICU acquired weakness 9. Ventilator-associated pneumonia 10. Bloodstream infection 11. Urinary tract infection
Notes	Email sent to Dr Yang January 17, 2022, and no reply was received. Original article was only available only in Chinese.

Outcome definitions from contributing trials

Co-primary outcomes

- Cognitive function measured by the Telephone Interview for Cognitive Status (TICS). TICS is an 11-item global mental status test. The individual item scores are summed to obtain the total score with higher scores meaning a better performance and a maximum score of 41. The total score provides a measure of global cognitive function.[19]
- HRQoL measured by the EuroQol five dimensions five levels questionnaire, including the visual analogue scale (EQ-VAS). EQ-VAS asks the patients to self-rate their perceived overall health on a scale from 0 ('the worst health you can imagine') to 100 ('the best health you can imagine').[20,21]

Exploratory outcomes

- Pulmonary function measured by the forced expired volume in the first second (FEV_1) and the forced vital capacity (FVC).
- Functional outcomes measured by the Barthel Index (BI),[22] the modified Rankin Scale (mRS),[23] the Glasgow Outcome Scale (GOS)[24] and its extended version (eGOS),[25] and the Cerebral Performance Category (CPC).[26] The BI is a 10-item scale of basic ADL. The 10 items focus on self-care and mobility, and the individual scores of the 10 items are summed to a maximum possible score of 100 (independent) and a minimum of 0 (totally dependent). The mRS is a global disability scale to measure the level of functional independence in daily activities. It consists of 7 grades going from 0 (no symptoms) to 6 (death). BI and mRS have mostly been used in stroke patients. The GOS is a 5-point scale going from 1 (death) to 5 (low disability). The eGOS is a 8-point scale, from 1 (death) to 8 (full recovery). GOS and its extended version have been mostly used in patients suffering from brain injury. The CPC is a five-point scale that ranges from 1 (good cerebral performance) to 5 (brain death), and it has been mostly used in survivors of cardiac arrest.
- Return to work defined as the employment status among survivors with a paid employment at randomisation.

Risk of bias

Two authors (EC and KUK) independently assessed risk of bias according to the Cochrane Handbook for Systematic Reviews of Interventions using the revised Risk of Bias tool 2. [27,28] This was done for each trial reporting at least one outcome of interest, and for each individual outcome. We assessed risk of bias in all the five mandatory domains: bias arising from the randomisation process; bias due to deviations from intended interventions; bias due to missing outcome data; bias in measurement of the outcome; and bias in the selection of the reported results. When assessing the domain ´bias due to deviations from intended interventions´, we judged the effect of assignment to the intervention (i.e., intention to treat effect). Each domain was adjudicated as 'low risk of bias', 'some concerns', or 'high risk of bias'. RCTs with 'low risk of bias' in all domains were classified as overall 'low risk of bias' trials. RCTs with one domain adjudicated at 'some concerns', but no domain adjudicated at 'high risk of bias', was classified as overall causing 'some concerns' of risk of bias. RCTs were classified as overall 'high risk of bias' trials if at least one domain was adjudicated at 'high risk of bias'. However, if a study was judged to have 'some concerns' of risk of bias for multiple domains, it may be judged as 'high risk of bias' overall if the assessors judged that the multiple concerns amount to a serious risk of bias.

Table S1 Risk of Bias for trials reporting TICS

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Mackle 2020[11]	Low	Low	High[1]	Low	Low	High

TICS: telephone interview for cognitive status. TICS is an 11-item global mental status test, with higher score meaning a better performance and a maximum score of 41.[19]

[1)] Neither an analysis correcting for missing outcome data nor a sensitivity analysis was performed. Health status of survivors could influence the rate of response. Lost to follow-up could have poorer or better outcome at follow-up than patients included in the final analysis.

Table S2 Risk of bias for trials reporting EQ-VAS

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Mackle 2020[11]	Low	Low	High[1]	Low	Low	High

EQ-VAS: EuroQol visual analogue scale score

EQ-VAS score ranges from 0 to 100, with higher scores indicating better health status.[20,21]

1) Neither an analysis correcting for missing outcome data nor a sensitivity analysis was performed. Health status of survivors could influence the rate of response (i.e. Lost to follow-up could have poorer or better outcome at follow-up than patients included in the final analysis).

Table S3 Risk of Bias for trials reporting FEV_1

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Gomersall 2002[6]	Low	High[1]	High[2]	Low	Some concerns[3]	High

FEV_1: forced expiratory volume in the first second

[1)] Post randomisation exclusion due to violation of inclusion criteria.

[2)] Neither an analysis correcting for missing outcome data nor a sensitivity analysis was performed. Health status of survivors could influence the rate of participation. Lost to follow-up could have poorer or better outcome at follow-up than patients included in the final analysis.

[3)] No protocol could be found.

Table S4 Risk of Bias for trials reporting FVC

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Gomersall 2002[6]	Low	High[1]	High[2]	Low	Some concerns[3]	High

FVC: forced vital capacity

[1)] Post randomisation exclusion due to violation of inclusion criteria.

[2)] Neither an analysis correcting for missing outcome data nor a sensitivity analysis was performed. Health status of survivors could influence the rate of participation. Lost to follow-up could have poorer or better outcome at follow-up than patients included in the final analysis.

[3)] No protocol could be found.

Table S5 Risk of Bias for trials reporting Barthel index

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Mazdeh 2015[13]	Some concerns[1]	High[2]	High[3]	Low	Some concerns[4]	High
Taher 2016[16]	Some concerns[5]	High[6]	High[7]	Low	Some concerns[8]	High

Barthel index: 10-item scale of the activities of daily living. The global score ranges from 0 (i.e. totally dependent) to 100 (i.e. independent) [23]

[1)] No information about the allocation sequence´s concealment.

[2)] No information on blinding and protocol adherence or oxygenation levels was found. A post-randomisation exclusion of patients who died or were lost to follow-up was performed.

[3)] Authors excluded both dead and patients lost to follow-up form the analysis and no further information about these was reported. Lost to follow-up patients could have poorer or better outcome than patients included in the final analysis.

[4)] The trial was retrospectively registered at the Iranian Registry of Clinical Trials (registration number: IRCT201212199647N2)

[5)] No information about the allocation sequence´s concealment.

[6)] A post-randomisation exclusion of patients who died or were lost to follow-up was performed.

[7)] Authors excluded both dead and patients lost to follow-up form the analysis and no further information about these was reported. Lost to follow-up patients could have poorer or better outcome than patients included in the final analysis.

[8)] The trial was registered at the Iranian Registry of Clinical Trials (ID number: IRCT201303209014N17) in recruitment phase.

Table S6 Risk of Bias for trials reporting modified Rankin Scale

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Mazdeh 2015[13]	Some concerns[1]	High[2]	High[3]	Low	High[4]	High
Taher 2016[16]	Some concerns[5]	High[6]	High[7]	Low	Some concerns[8]	High

The modified Rankin Scale is a seven-grade disability scale going from 0 (i.e. no symptoms) to 6 (i.e. death).[24]

1) No information about the allocation sequence´s concealment.

2) No information on blinding and protocol adherence or oxygenation levels was found. A post-randomisation exclusion of patients who died or were lost to follow-up was performed.

3) Authors excluded both dead and patients lost to follow-up form the analysis and no further information about these was reported. Lost to follow-up patients could have poorer or better outcome than patients included in the final analysis.

4) The trial was retrospectively registered at the Iranian Registry of Clinical Trials (registration number: IRCT201212199647N2). Two analyses of the outcome were performed but the general linear model analysis was not fully reported.

5) No information about the allocation sequence´s concealment.

6) A post-randomisation exclusion of patients who died or were lost to follow-up was performed.

7) Authors excluded both dead and patients lost to follow-up form the analysis and no further information about these was reported. Lost to follow-up patients could have poorer or better outcome than patients included in the final analysis.

8) The trial was registered at the Iranian Registry of Clinical Trials (ID number: IRCT201303209014N17) in recruitment phase.

Table S7 Risk of Bias for trials reporting GOS or extended GOS

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Lång 2018[10]	Low	High[1]	High[2]	Low	Low	High
Mackle 2020[11]	Low	Low	High[3]	Low	Some concerns[4]	High
Taher 2016[16]	Some concerns[5]	High[6]	High[7]	Low	Some concerns[8]	High

GOS: Glasgow Outcome Scale; eGOS: extended Glasgow Outcome Scale

The GOS is a 5-point scale ranging from 1 (i.e. death) to 5 (i.e. low disability).[25] The extended version of GOS is a 8-point scale, from 1 being death to 8 meaning Upper good recovery (i.e. full recovery).[26]

[1)] 5 patients were excluded from the analysis due to loss-to follow-up, not specified group allocation.

[2)] Neither an analysis correcting for missing outcome data nor a sensitivity analysis was performed. Health status of survivors could influence the rate of response. Lost to follow-up could have poorer or better outcome at follow-up than patients included in the final analysis.

[3)] Neither an analysis correcting for missing outcome data nor a sensitivity analysis was performed. Health status of survivors could influence the rate of response. Lost to follow-up could have poorer or better outcome at follow-up than patients included in the final analysis.

[4)] Post-hoc analysis

[5)] No information about the allocation sequence´s concealment.

[6)] A post-randomisation exclusion of patients who died or were lost to follow-up was performed.

[7)] Authors excluded both dead and patients lost to follow-up form the analysis and no further information about these was reported. Lost to follow-up patients could have poorer or better outcome than patients included in the final analysis.

[8)] The trial was registered at the Iranian Registry of Clinical Trials (ID number: IRCT201303209014N17) in recruitment phase.

Table S8 Risk of Bias for trial reporting CPC

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Jakkula 2018[8]	Low	Low	Low	Low	Low	Low

CPC: Cerebral Performance Category

CPC is a five-point scale that ranges from 1 (i.e. good cerebral performance) to 5 (i.e. brain death).[27]

Table S9 Risk of Bias for trials reporting return to work

Study	Randomisation process	Deviations from the intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Risk of Bias
Mackle 2019[11]	Low	Low	Low	Low	Low	Low

Table S10 CDIM for trials reporting Barthel index

	Mazdeh 2014[13]	Taher 2016[16]	Score
Setting 1. Years reported (A), performed in developed vs developing country (B), unit type (C)	Publication year 2014 (A), Developing country (Iran) (B), ICU (C)	Conduct 2014 (publication year 2015) (A) Developing country (Iran) (B) ICU (C)	0
Population 2. Age	Not reported	43	1
3. Sex	45% women	29% women	1
4. Participant inclusion criteria and baseline disease severity	Ischemic and hemorragic stroke with no need for surgical intervention, 40≤ age≤70, GCS>12, patients with isolated brain damage and intact airway control, less than 12 hours have passed since the accident; 7≤ NIHSS≤9	TBI, 18≤ age≤65, hemodynamic stability, 3≤GCS≤8, less than 6 hours have passed since the accident; baseline GCS 7.4	2
5. Comorbidites	Not reported	Not reported	1
Intervention 6. Intensity, strengths, or duration of intervention	FiO_2 of 0.50 - oxygen therapy with Venturi mask for the first 12 hours	FiO_2 0.80 through mechanical ventilation for 6 hours	2
7. Timing	Intervention started at ICU admission; less than 12 hours passed since the accident	Intervention started at ICU admission; less than 6 hours passed since the accident	1
8. Control intervention	Not supplemental oxygen	FiO_2 0.50 through mechanical ventilation	2
9. Cointerventions	Not specified cointerventions	Not specified cointerventions	2*
Outcome 10. Definition of the outcome in the meta-analysis	Barthel index	Barthel index	0
11. Timing of outcome measurement	6 months	6 months	0
Total score			12

CDIM: clinical diversity in meta-analyses; ICU: intensive care unit; GCS: Glasgow coma scale (ref); NIHSS: National institutes of health stroke scale(ref); FiO_2: fraction of inspired oxygen

*given heterogeneity between the included populations, we expected that standard of care would differ.

Table S11 CDIM for trials reporting mRS

	Mazdeh 2014[13]	Taher 2016[16]	Score
Setting 1. Years reported (A), performed in developed vs developing country (B), unit type (C)	Publication year 2014 (A), Developing country (Iran) (B), ICU (C)	Conduct 2014 (publication year 2015) (A) Developing country (Iran) (B) ICU (C)	0
Population 2. Age	Not reported	43	1
3. Sex	45% women	29% women	1
4. Participant inclusion criteria and baseline disease severity	Ischemic and hemorragic stroke with no need for surgical intervention, 40≤ age≤70, GCS>12, patients with isolated brain damage and intact airway control, less than 12 hours have passed since the accident; 7≤ NIHSS≤9	TBI, 18≤ age≤65, hemodynamic stability, 3≤GCS≤8, less than 6 hours have passed since the accident; baseline GCS 7.4	2
5. Comorbidities	Not reported	Not reported	1
Intervention 6. Intensity, strengths, or duration of intervention	FiO_2 of 0.50 - oxygen therapy with Venturi mask for the first 12 hours	FiO_2 0.80 through mechanical ventilation for 6 hours	2
7. Timing	Intervention started at ICU admission; less than 12 hours passed since the accident	Intervention started at ICU admission; less than 6 hours passed since the accident	1
8. Control intervention	Not supplemental oxygen	FiO_2 0.50 through mechanical ventilation	2
9. Cointerventions	Not specified cointerventions	Not specified cointerventions	2*
Outcome 10. Definition of the outcome in the meta-analysis	mRS	mRS	0
11. Timing of outcome measurement	6 months	6 months	0
Total score			12

CDIM: clinical diversity in meta-analyses; mRS: modified Rankin scale (ref); ICU: intensive care unit; GCS: Glasgow coma scale (ref); NIHSS: National institutes of health stroke scale(ref); FiO_2: fraction of inspired oxygen

*given heterogeneity between the included populations, we expected that standard of care would differ.

Table S12 Raw data for extended Glasgow Outcome Scale

A: Date from Lång et al.[10]*

Extended Glasgow Outcome Scale – No. of patients	Lower Oxygenation Group (N=26)	Higher Oxygenation Group (N=37)
Score 1 - "Death"	8	9
Score 2 – "Persistent vegetative state"	0	1
Score 3 – "Lower severe disability"	4	8
Score 4 – "Upper severe disability"	2	4
Score 5 – "Lower moderate disability"	3	2
Score 6 – "Upper moderate disability"	6	5
Score 7 – "Lower good recovery"	2	4
Score 8 – "Upper good recovery"	1	4

B: Data from Mackle et al.[11]*

Extended Glasgow Outcome Scale – No. of patients	Lower Oxygenation Group (N=166)	Higher Oxygenation Group (N=157)
Score 1 - "Death"	75	74
Score 2 – "Persistent vegetative state"	0	0
Score 3 – "Lower severe disability"	16	16
Score 4 – "Upper severe disability"	10	10
Score 5 – "Lower moderate disability"	10	13
Score 6 – "Upper moderate disability"	21	17
Score 7 – "Lower good recovery"	12	13
Score 8 – "Upper good recovery"	22	14

C: Data from Taher et al.[16]

Extended Glasgow Outcome Scale – No. of patients	Lower Oxygenation Group (N=34)	Higher Oxygenation Group (N=34)
Score 1 - "Death"	0	0
Score 2 – "Persistent vegetative state"	0	0
Score 3 – "Severe disability"	3	0
Score 4 – "Moderate disability"	16	9
Score 5 – "Low disability"	15	25

*The presented data were collected upon contacting the corresponding authors.

Table S13 CDIM in trials evaluating GOS or extended GOS

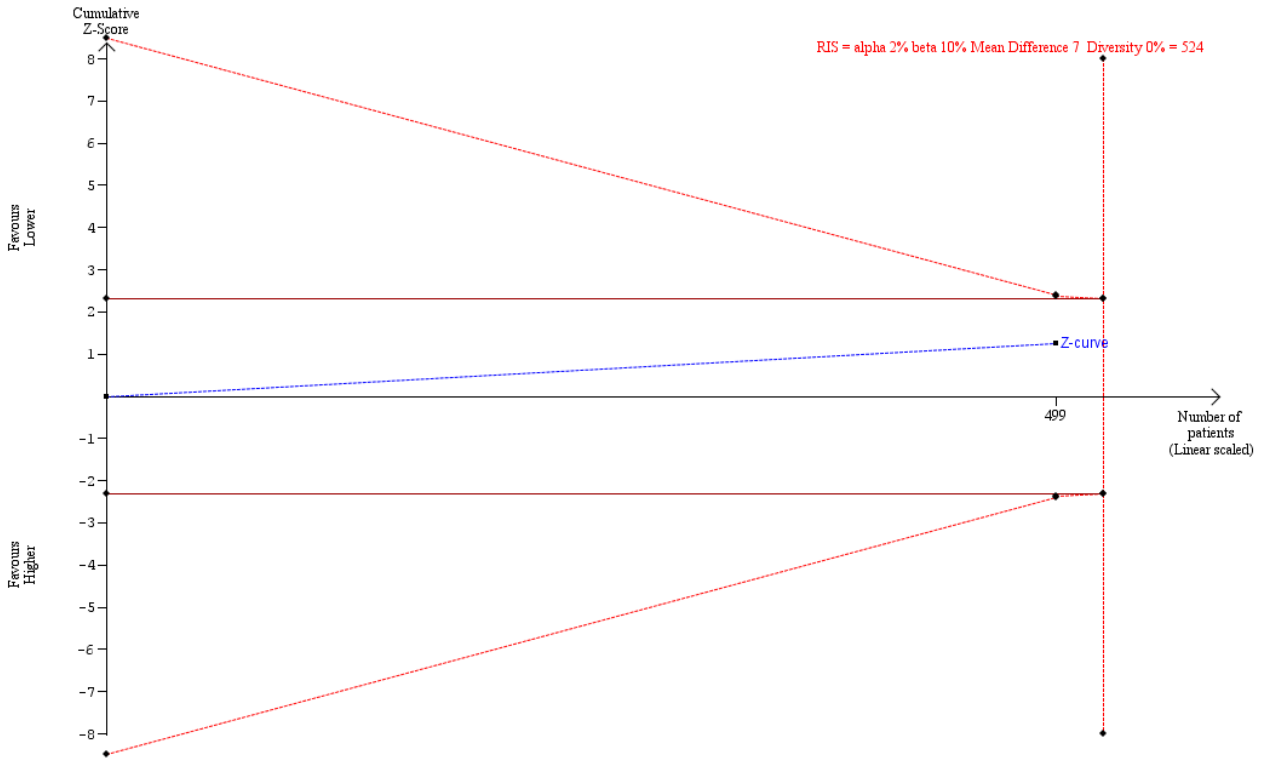
	Taher 2016[16]	Lång 2018[10]	Mackle 2020[11]	Score
Setting 1. Years reported (A), performed in developed vs developing country (B), unit type (C)	Conduct 2014 (publication year 2015) (A), Developing country (Iran) (B), ICU (C)	Conduct 2010-2013 (publication year 2018) (A), Developed countries (Finland) (B), ICU (C)	Conduct 2015-2018 (publication year 2020) (A), Developed country (Australia and New Zealand) (B), ICU (C)	1
Population 2. Age	43	44	58	1
3. Sex	29% women	17% women	37% women	1
4. Participant inclusion criteria and baseline disease severity	TBI, 18≤ age≤65, hemodynamic stability, 3≤GCS≤8, less than 6 hours passed since the accident; baseline GCS 7.4	Isolated non-penetrating TBI or multiple trauma with TBI and GCS ≤8, expected need for mechanical ventilation >24h, recruitment within 18h within ICU admission; less than 36 h since the event; baseline SOFA score 10 APACHE II score	All adults (≥18 years old) expecting to receive mechanical ventilation in the ICU beyond the day after recruitment; APACHE II score 23,5	2
5. Comorbidites	Not reported	22.5 Heart disease (20%); Neurological disease (4.6%); Pulmonary disease (9.2%); Diabetes (23.1%); Anticoagulation therapy (3.1%)	Not reported	1
Intervention 6. Intensity, strengths, or duration of intervention	FiO_2 0.80 through mechanical ventilation for 6 hours	FiO_2 0.70 through mechanical ventilation for a maximum time of 14 days	SpO_2 90-100%, FiO_2<0.30 discouraged, through mechanical ventilation, throughout ICU admission to a maximum of 28 days	2
7. Timing	Intervention started at ICU admission	Recruitment within 18h within ICU admission	Recruitment within 2 hours since mechanical	1

			ventilation was started	
8. Control intervention	FiO_2 0.50 through mechanical ventilation	FiO_2 0.40 through mechanical ventilation	SpO_2 90-96%	2
9. Cointerventions	Not specified cointerventions	Not specified cointerventions	Not specified cointerventions	2*
Outcome 10. Definition of the outcome in the meta-analysis	GOS	Extended GOS	Extended GOS	1
11. Timing of outcome measurement	6 months	6 months	180 days	0
Total score				14

CDIM: clinical diversity in meta-analyses; GOS: Glasgow outcome scale (ref); ICU: intensive care unit; TBI: traumatic brain injury; SOFA: Sequential Organ Failure assessment(ref); APACHE II: Acute Physiology and Chronic Health Evaluation II (ref); FiO_2: fraction of inspired oxygen; SpO_2: peripheral oxygen saturation

*Given heterogeneity between the included populations, we expected that standard of care would differ.

Figure S1 Trial Sequential Analysis (TSA) of HRQoL measured as EQ-VAS



HRQoL denotes health-related quality of life; EQ-VAS: EuroQol visual analogue scale. TSA demonstrated that with an anticipated mean difference of 7 points, a type 1 error level of 2%, a type 2 error level of 10%, and a diversity of 0%, the required information size (RIS) was 524 participants. The cumulative Z-curve did not cross any boundaries for benefit and harm, nor the trial sequential monitoring boundaries for futility.

References

1. Page MJ, McKenzie JE, Bossuyt PM et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021; 372: n71
2. Asfar P, Schortgen F, Boisramé-Helms J et al. Hyperoxia and hypertonic saline in patients with septic shock (HYPERS2S): a two-by-two factorial, multicentre, randomised, clinical trial. Lancet Respir Med 2017; 5: 180-90
3. Barrot L, Asfar P, Mauny F et al. Liberal or Conservative Oxygen Therapy for Acute Respiratory Distress Syndrome. N Engl J Med 2020; 382: 999-1008
4. Gelissen H, de Grooth HJ, Smulders Y et al. Effect of Low-Normal vs High-Normal Oxygenation Targets on Organ Dysfunction in Critically Ill Patients: A Randomized Clinical Trial. Jama 2021
5. Girardis M, Busani S, Damiani E et al. Effect of Conservative vs Conventional Oxygen Therapy on Mortality Among Patients in an Intensive Care Unit: The Oxygen-ICU Randomized Clinical Trial. Jama 2016; 316: 1583-89
6. Gomersall CD, Joynt GM, Freebairn RC et al. Oxygen therapy for hypercapnic patients with chronic obstructive pulmonary disease and acute respiratory failure: a randomized, controlled pilot study. Crit Care Med 2002; 30: 113-6
7. Ishii K, Morimatsu H, Hyodo T et al. Relationship between inspired oxygen concentration and atelectasis formation after extubation. Critical Care Medicine 2018; 46: 533
8. Jakkula P, Reinikainen M, Hästbacka J et al. Targeting two different levels of both arterial carbon dioxide and arterial oxygen after cardiac arrest and resuscitation: a randomised pilot trial. Intensive Care Med 2018; 44: 2112-21
9. Jun J, Sun L, Wang Y et al. Invasive mechanical ventilation with high concentration oxygen therapy for AECOPD patients with acute myocardial infarction. Chest 2019; 156: A958
10. Lång M, Skrifvars MB, Siironen J et al. A pilot study of hyperoxemia on neurological injury, inflammation and oxidative stress. Acta Anaesthesiol Scand 2018; 62: 801-10
11. Mackle D, Bellomo R, Bailey M et al. Conservative Oxygen Therapy during Mechanical Ventilation in the ICU. N Engl J Med 2020; 382: 989-98
12. Martin DS, McNeil M, Brew-Graves C et al. A feasibility randomised controlled trial of targeted oxygen therapy in mechanically ventilated critically ill patients. J Intensive Care Soc 2021; 22: 280-87
13. Mazdeh M, Taher A, Torabian S et al. Effects of Normobaric Hyperoxia in Severe Acute Stroke: a Randomized Controlled Clinical Trial Study. Acta Med Iran 2015; 53: 676-80
14. Panwar R, Hardie M, Bellomo R et al. Conservative versus Liberal Oxygenation Targets for Mechanically Ventilated Patients. A Pilot Multicenter Randomized Controlled Trial. Am J Respir Crit Care Med 2016; 193: 43-51
15. Schjørring OL, Klitgaard TL, Perner A et al. Lower or Higher Oxygenation Targets for Acute Hypoxemic Respiratory Failure. N Engl J Med 2021; 384: 1301-11
16. Taher A, Pilehvari Z, Poorolajal J et al. Effects of Normobaric Hyperoxia in Traumatic Brain Injury: A Randomized Controlled Clinical Trial. Trauma Mon 2016; 21: e26772
17. Yang X, Shang Y, Yuan S. Low versus high pulse oxygen saturation directed oxygen therapy in critically ill patients: a randomized controlled pilot study. J Thorac Dis 2019; 11: 4234-40
18. Yang W, Zhang L. [Observation of the curative effect of conservative oxygen therapy in mechanical ventilation of patients with severe pneumonia]. Zhonghua Wei Zhong Bing Ji Jiu Yi Xue 2021; 33: 1069-73
19. Brandt J, Spencer M, Folstein M. The telephone interview for cognitive status. Neuropsychiatry, Neuropsychology and Behavioral Neurology 1988; 1: 111-17
20. Janssen MF, Bonsel GJ, Luo N. Is EQ-5D-5L Better Than EQ-5D-3L? A Head-to-Head Comparison of Descriptive Systems and Value Sets from Seven Countries. Pharmacoeconomics 2018; 36: 675-97

21. Herdman M, Gudex C, Lloyd A et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). Qual Life Res 2011; 20: 1727-36
22. Mahoney FI, Barthel DW. Functional evaluation: The Barthel Index. Md State Med J 1965; 14: 61-5
23. van Swieten JC, Koudstaal PJ, Visser MC et al. Interobserver agreement for the assessment of handicap in stroke patients. Stroke 1988; 19: 604-7
24. Jennett B, Bond M. Assessment of outcome after severe brain damage. Lancet 1975; 1: 480-4
25. Jennett B, Snoek J, Bond MR et al. Disability after severe head injury: observations on the use of the Glasgow Outcome Scale. J Neurol Neurosurg Psychiatry 1981; 44: 285-93
26. Jacobs I, Nadkarni V, Bahr J et al. Cardiac arrest and cardiopulmonary resuscitation outcome reports: update and simplification of the Utstein templates for resuscitation registries: a statement for healthcare professionals from a task force of the International Liaison Committee on Resuscitation. Circulation 2004; 110: 3385-97
27. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.2 (updated February 2021). Cochrane, 2021. Available from www.training.cochrane.org/handbook.
28. Sterne JAC, Savović J, Page MJ et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. BMJ 2019; 366: l4898.